HVAC SYSTEM

COMPLETION & COMMISSIONING

Project Address: ______________________________

Project Name: ______________________________

Permit Number: ______________________________

HVAC Contractor: ______________________________

License Number: ______________________________

Contact Phone: ______________________________

PURPOSE: This document verifies proper installation, startup, and commissioning of the complete HVAC system. Proper commissioning ensures the system operates efficiently, safely, and as designed.

EQUIPMENT INSTALLATION

Indoor Unit (Furnace/Air Handler):

Make/Model: ______________________________

Serial Number: ______________________________

BTU Input/Output: ______________________________

Fuel Type: ■ Natural Gas ■ Propane ■ Electric ■ Oil

AFUE Rating: ______________________________

Location: ______________________________

Installation Date: ______________________________

- ■ ■ Unit installed level and properly supported
- ■ ■ Proper clearances maintained (front, sides, top per manufacturer)
- ■ ■ Unit properly secured to platform or hung from joists
- ■ ■ Gas line connected properly (if gas) - no leaks
- ■ ■ Gas shut-off valve within 6 feet (if gas)
- ■ ■ Gas line pressure tested (if gas)
- ■ ■ Electrical connection made - proper voltage and amperage
- ■ ■ Electrical disconnect within sight of unit
- ■ ■ Condensate drain line installed with proper slope
- ■ ■ Condensate drain terminates appropriately
- ■ ■ Condensate trap installed (if required)
- ■ ■ Secondary drain pan installed (if in attic/living space)
- ■ ■ Secondary drain pan alarm/sensor installed
- ■ ■ Combustion air intake adequate (if gas)
- ■ ■ Vent pipe properly installed (if gas)
- ■ ■ Vent pipe properly supported and sloped
- ■ ■ Vent termination proper distance from windows/openings
- ■ ■ Filter access accessible
- ■ ■ Blower door/access panels properly secured

Outdoor Unit (Condensing Unit/Heat Pump):

Make/Model: ____________________

Serial Number: ____________________

Tonnage: ____________________

SEER Rating: ____________________

Type: ■ Air Conditioner ■ Heat Pump

Refrigerant Type: ____________________

Location: ____________________

- ■ Unit installed on level pad (concrete or composite)
- ■ Pad stable and properly sized
- ■ Unit level side-to-side and front-to-back (within 1/4")
- ■ Proper clearances maintained (12" sides, 24" service side, 60" top)
- ■ No vegetation or obstructions blocking airflow
- ■ Refrigerant lines connected and insulated
- ■ Line set insulation complete - no exposed copper
- ■ Line set properly secured to structure
- ■ Electrical disconnect installed within sight of unit
- ■ Electrical disconnect proper rating (check data plate)
- ■ Whip (flexible conduit) to unit installed properly
- ■ Electrical connections tight at contactor and terminals
- ■ Condenser fan rotates freely by hand (power off)
- ■ Coil fins undamaged and clean
- ■ Service valves accessible

DUCTWORK COMPLETION

- ■ All supply registers installed
- ■ All supply registers properly secured
- ■ Register boot connections sealed with mastic
- ■ All return grilles installed
- ■ Return grille connections sealed
- ■ Main trunk line connections sealed
- ■ All branch take-offs sealed
- ■ Flexible duct properly supported (every 4-5 feet)
- ■ Flexible duct not kinked or crushed
- ■ Ductwork insulation complete (if required)
- ■ No disconnected ductwork
- ■ Filter installed - correct size noted: _______________
- ■ Filter access accessible
- ■ Return air pathways adequate (no blocked returns)
- ■ Access panels installed for dampers and controls

Register/Grille Schedule:

Room	Supply Register Size	Qty	Return Grille Size	Qty	Notes

Room	Supply Register Size	Qty	Return Grille Size	Qty	Notes

THERMOSTAT INSTALLATION

Make/Model: ______________________________

Type: ■ Programmable ■ Non-Programmable ■ Smart/WiFi

Location: ______________________________

Number of Zones: ______________________________

- ■ ■ Thermostat located on interior wall (not exterior)
- ■ ■ Location away from direct sunlight
- ■ ■ Location away from heat sources (lamps, appliances)
- ■ ■ Location away from drafts (doors, windows)
- ■ ■ Thermostat mounted level at 52-60" height
- ■ ■ All wires properly labeled
- ■ ■ All wire connections tight
- ■ ■ Thermostat wired correctly - tested with system
- ■ ■ Programmable thermostat programmed (if applicable)
- ■ ■ Smart thermostat connected to WiFi (if applicable)
- ■ ■ Homeowner trained on thermostat operation
- ■ ■ Manual/instructions left with homeowner

SYSTEM COMMISSIONING

Professional commissioning must be performed by qualified HVAC technician:

Commissioning Technician: ______________________________

Company: ______________________________

License Number: ______________________________

Commissioning Date: ______________________________

Refrigerant Charging (Air Conditioner/Heat Pump):

Refrigerant Type: ______________________________

Charge Weight (lbs): ______________________________

Charging Method: ■ Subcooling ■ Superheat ■ Weigh-In

Outdoor Temperature (°F): ______________________________

Indoor Temperature (°F): ______________________________

Indoor Relative Humidity (%): ______________________________

Subcooling Method (preferred for TXV systems):

High Side Pressure (PSI): ______________________________

Saturation Temperature (°F): ______________________________

Liquid Line Temperature (°F): ______________________________

Subcooling (°F): ______________________________

Target Subcooling per Manufacturer: ______________________________

Subcooling Within Spec: ■ Yes ■ No

Superheat Method (for fixed orifice/piston systems):

Low Side Pressure (PSI): ______________________________

Saturation Temperature (°F): ______________________________

Suction Line Temperature (°F): ______________________________

Superheat (°F): ____________________

Target Superheat per Manufacturer: ____________________

Superheat Within Spec: ■ Yes ■ No

Airflow Verification:

Blower Speed Setting: ______________________

Supply Air Temperature (°F): ______________________

Return Air Temperature (°F): ______________________

Temperature Split (°F): ______________________

Target Split (cooling: 15-20°F): ______________________

Airflow CFM (if measured): ______________________

Target Airflow (400 CFM/ton): ______________________

Airflow Adequate: ■ Yes ■ No

Gas Furnace Checks (if applicable):

Gas Type: ■ Natural Gas ■ Propane

Manifold Pressure (in. w.c.): ______________________

Target Manifold Pressure: ______________________

Temperature Rise (°F): ______________________

Target Temperature Rise Range: ______________________

Flame Appearance: ■ Blue/Stable ■ Other: ________

Carbon Monoxide Test (PPM): ______________________

CO Level Acceptable (<10 PPM): ■ Yes ■ No

Vent Draft Adequate: ■ Yes ■ No ■ N/A

OPERATIONAL TESTING

Heating Mode Test:

- ■ Thermostat set to HEAT mode
- ■ Temperature set above room temperature
- ■ Furnace/heat pump starts within 1-2 minutes
- ■ Ignition sequence normal (if gas)
- ■ Burner flames stable and blue (if gas)
- ■ Blower starts after heat-up delay
- ■ Warm air from all supply registers
- ■ Airflow adequate at all registers
- ■ System runs continuously without cycling
- ■ No unusual noises during operation
- ■ System reaches temperature setpoint and cycles off
- ■ Blower continues for cool-down period
- ■ No error codes or warning lights

Cooling Mode Test:

- ■ Thermostat set to COOL mode
- ■ Temperature set below room temperature
- ■ Outdoor unit starts (compressor and fan)
- ■ Indoor blower starts
- ■ Cool air from all supply registers
- ■ Temperature drop 15-20°F (supply vs. return)
- ■ Condensate draining properly
- ■ No water leaks from indoor unit
- ■ Outdoor unit running smoothly - no unusual noises
- ■ System runs continuously without short-cycling
- ■ System reaches temperature setpoint and cycles off
- ■ No error codes or warning lights

General System Operation:

- ■ Thermostat controls system properly
- ■ Fan mode AUTO and ON both work
- ■ System cycles properly - not short-cycling
- ■ No vibration in ductwork during operation
- ■ No whistling or whooshing sounds from registers
- ■ All rooms receiving adequate airflow
- ■ Return air adequate - no negative pressure issues
- ■ Emergency heat operates (if heat pump) - ■ N/A
- ■ Reversing valve operates (if heat pump) - ■ N/A
- ■ Outdoor unit clean and free of debris

AIRFLOW BALANCING

Verify adequate airflow to all rooms:

Room	Supply Register	Airflow	Temperature	Damper Adjustment	Notes
		■ Good ■ Low			
		■ Good ■ Low			
		■ Good ■ Low			
		■ Good ■ Low			
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		■ Good ■ Low			
		■ Good ■ Low			

- ■ All rooms receiving airflow
- ■ Hot/cold spots identified and addressed
- ■ Dampers adjusted for balanced airflow
- ■ Temperature variation between rooms minimal (within 3-4°F)

FINAL HVAC INSPECTION

Inspection Requested Date: ______________________________

Inspection Scheduled Date: ______________________________

Inspector Name: ______________________________

Inspection Date: ______________________________

Inspection Time: ______________________________

Pre-Inspection Checklist:

- ■ All equipment properly installed
- ■ All ductwork complete and sealed
- ■ System commissioned by qualified technician
- ■ Refrigerant charge verified
- ■ Heating and cooling both tested
- ■ Thermostat functioning properly
- ■ All required disconnects installed
- ■ Proper clearances maintained
- ■ Condensate drains functioning
- ■ Work completed per approved plans

Inspection Result:

■ PASSED - Certificate Issued

■ FAILED - Corrections Required (see below)

Required Corrections:

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____

Corrections Completed Date: ____________________

Re-Inspection Date: ____________________

Re-Inspection Result: ■ PASSED ■ FAILED

WARRANTY & DOCUMENTATION

Equipment Warranty Period: ______________________________

Parts Warranty Period: ______________________________

Labor Warranty Period: ______________________________

Compressor Warranty Period: ______________________________

Registration Required: ■ Yes ■ No

Registration Completed: ■ Yes ■ No ■ N/A

Registration Date: ______________________________

Documentation Provided to Homeowner:

- ■ ■ Equipment owner's manuals (indoor and outdoor units)
- ■ ■ Thermostat manual and programming guide
- ■ ■ Warranty information and registration
- ■ ■ Filter size and replacement schedule
- ■ ■ Maintenance schedule and recommendations
- ■ ■ Contractor contact information for service
- ■ ■ As-built duct layout (if available)
- ■ ■ Commissioning report with all measurements

HVAC Technician Signature: ______________________________

Date: ______________________________

Owner/Builder Signature: ______________________________

Date: ______________________________